

WORKSHEET-2 OUTPUTS

```
   age      job marital education default balance housing loan \
0   58  management  married   tertiary      no      2143    yes   no
1   44   technician   single  secondary      no        29    yes   no
2   33  entrepreneur  married  secondary      no         2    yes  yes
3   47  blue-collar  married   unknown      no     1506    yes   no
4   33      unknown   single   unknown      no         1     no   no
```

```
   contact  day month duration campaign  pdays  previous  poutcome   y
0  unknown    5  may      261         1     -1         0  unknown  no
1  unknown    5  may      151         1     -1         0  unknown  no
2  unknown    5  may       76         1     -1         0  unknown  no
3  unknown    5  may       92         1     -1         0  unknown  no
4  unknown    5  may      198         1     -1         0  unknown  no
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 45211 entries, 0 to 45210
```

```
Data columns (total 17 columns):
```

```
#   Column      Non-Null Count  Dtype
---  -
0   age        45211 non-null    int64
1   job         45211 non-null    object
2   marital     45211 non-null    object
3   education   45211 non-null    object
4   default     45211 non-null    object
5   balance     45211 non-null    int64
6   housing     45211 non-null    object
7   loan        45211 non-null    object
8   contact     45211 non-null    object
9   day         45211 non-null    int64
10  month       45211 non-null    object
11  duration    45211 non-null    int64
12  campaign    45211 non-null    int64
13  pdays      45211 non-null    int64
14  previous    45211 non-null    int64
15  poutcome    45211 non-null    object
16  y           45211 non-null    object
```

```
dtypes: int64(7), object(10)
```

```
memory usage: 5.9+ MB
```

```
None
```

```

Index(['job', 'marital', 'education', 'default', 'housing', 'loan', 'contact',
      'month', 'poutcome', 'y'],
      dtype='object')
['management' 'technician' 'entrepreneur' 'blue-collar' 'unknown'
 'retired' 'admin.' 'services' 'self-employed' 'unemployed' 'housemaid'
 'student']
['married' 'single' 'divorced']
['tertiary' 'secondary' 'unknown' 'primary']
['no' 'yes']
['yes' 'no']
['no' 'yes']
['unknown' 'cellular' 'telephone']
['may' 'jun' 'jul' 'aug' 'oct' 'nov' 'dec' 'jan' 'feb' 'mar' 'apr' 'sep']
['unknown' 'failure' 'other' 'success']
['no' 'yes']
age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays      0
previous     0
poutcome     0
y            0
dtype: int64

```

	age	balance	day	duration	campaign	pdays	previous	
0	58	2143	5	261	1	-1	0	
1	44	29	5	151	1	-1	0	
2	33	2	5	76	1	-1	0	
3	47	1506	5	92	1	-1	0	
4	33	1	5	198	1	-1	0	
	age	balance	day	duration	campaign	pdays	previous	\
count	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	
mean	40.936210	1362.272058	15.806419	258.163080	2.763841			
std	10.618762	3044.765829	8.322476	257.527812	3.098021			
min	18.000000	-8019.000000	1.000000	0.000000	1.000000			
25%	33.000000	72.000000	8.000000	103.000000	1.000000			
50%	39.000000	448.000000	16.000000	180.000000	2.000000			
75%	48.000000	1428.000000	21.000000	319.000000	3.000000			
max	95.000000	102127.000000	31.000000	4918.000000	63.000000			
	pdays	previous						
count	45211.000000	45211.000000						
mean	40.197828	0.580323						
std	100.128746	2.303441						
min	-1.000000	0.000000						
25%	-1.000000	0.000000						
50%	-1.000000	0.000000						
75%	-1.000000	0.000000						
max	871.000000	275.000000						

First 5 rows:

	Student ID	Age	Gender	Height	Weight	Blood Type	BMI
0	1.0	18.0	Female	161.777924	72.354947	0	27.645835
1	2.0	NaN	Male	152.069157	47.630941	B	NaN
2	3.0	32.0	Female	182.537664	55.741083	A	16.729017
3	NaN	30.0	Male	182.112867	63.332207	B	19.096042
4	5.0	23.0	Female	NaN	46.234173	0	NaN

	Temperature	Heart Rate	Blood Pressure	Cholesterol	Diabetes	Smoking
0	NaN	95.0	109.0	203.0	No	NaN
1	98.714977	93.0	104.0	163.0	No	No
2	98.260293	76.0	130.0	216.0	Yes	No
3	98.839605	99.0	112.0	141.0	No	Yes
4	98.480008	95.0	NaN	231.0	No	No

DataFrame Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200000 entries, 0 to 199999

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Student ID	180000 non-null	float64
1	Age	180000 non-null	float64
2	Gender	180000 non-null	object
3	Height	180000 non-null	float64
4	Weight	180000 non-null	float64
5	Blood Type	180000 non-null	object
6	BMI	180000 non-null	float64
7	Temperature	180000 non-null	float64
8	Heart Rate	180000 non-null	float64
9	Blood Pressure	180000 non-null	float64
10	Cholesterol	180000 non-null	float64
11	Diabetes	180000 non-null	object
12	Smoking	180000 non-null	object

dtypes: float64(9), object(4)

memory usage: 19.8+ MB

None

Missing values per column:

Student ID	20000
Age	20000
Gender	20000
Height	20000
Weight	20000
Blood Type	20000
BMI	20000
Temperature	20000
Heart Rate	20000
Blood Pressure	20000
Cholesterol	20000
Diabetes	20000
Smoking	20000

dtype: int64

Filled missing values in numeric column 'Student ID' with mean.

Filled missing values in numeric column 'Age' with mean.

Filled missing values in categorical column 'Gender' with mode.

Filled missing values in numeric column 'Height' with mean.

Filled missing values in numeric column 'Weight' with mean.

Filled missing values in categorical column 'Blood Type' with mode.

Filled missing values in numeric column 'BMI' with mean.

Filled missing values in numeric column 'Temperature' with mean.

Filled missing values in numeric column 'Heart Rate' with mean.

Filled missing values in numeric column 'Blood Pressure' with mean.

Filled missing values in numeric column 'Cholesterol' with mean.

Filled missing values in categorical column 'Diabetes' with mode.

Filled missing values in categorical column 'Smoking' with mode.

Missing values after imputation:

```
Student ID      0
Age             0
Gender          0
Height         0
Weight         0
Blood Type     0
BMI            0
Temperature    0
Heart Rate     0
Blood Pressure 0
Cholesterol    0
Diabetes       0
Smoking        0
dtype: int64
```

Number of duplicate rows: 12572

DataFrame after removing duplicates:

	Student ID	Age	Gender	Height	Weight	Blood Type	\
0	1.000000	18.000000	Female	161.777924	72.354947	0	
1	2.000000	26.021561	Male	152.069157	47.630941	B	
2	3.000000	32.000000	Female	182.537664	55.741083	A	
3	49974.042078	30.000000	Male	182.112867	63.332207	B	
4	5.000000	23.000000	Female	174.947103	46.234173	0	

	BMI	Temperature	Heart Rate	Blood Pressure	Cholesterol	Diabetes	\
0	27.645835	98.600948	95.0	109.000000	203.0	No	
1	23.338869	98.714977	93.0	104.000000	163.0	No	
2	16.729017	98.260293	76.0	130.000000	216.0	Yes	
3	19.096042	98.839605	99.0	112.000000	141.0	No	
4	23.338869	98.480008	95.0	114.558033	231.0	No	

	Smoking
0	No
1	No
2	No
3	Yes
4	No

```
Mean Fare: 84.1546875
Median Fare: 60.287499999999994
Maximum Fare: 512.3292
Minimum Fare: 0.0
```

Number of null values in 'Age': 177

DataFrame after dropping rows with null Age:

	Name	Pclass	Sex	Age	\
0	Braund, Mr. Owen Harris	3	male	22.0	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	1	female	38.0	
2	Heikkinen, Miss. Laina	3	female	26.0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	1	female	35.0	
4	Allen, Mr. William Henry	3	male	35.0	

	Fare	Survived
0	7.2500	0
1	71.2833	1
2	7.9250	1
3	53.1000	1
4	8.0500	0

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked_C	Embarked_Q	Embarked_S
0	0	A/5 21171	7.2500	NaN	False	False	True
1	0	PC 17599	71.2833	C85	True	False	False
2	0	STON/O2. 3101282	7.9250	NaN	False	False	True
3	0	113803	53.1000	C123	False	False	True
4	0	373450	8.0500	NaN	False	False	True

Mean survival rates by gender:

Sex

female 0.742038

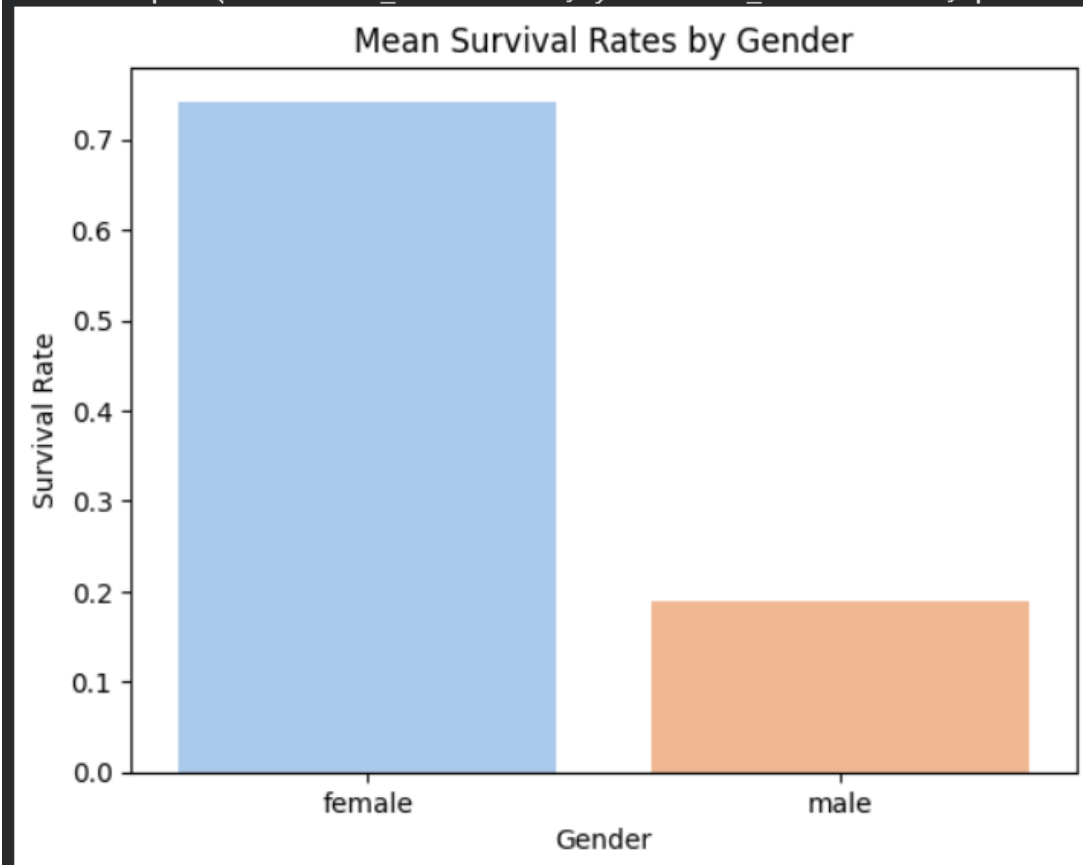
male 0.188908

Name: Survived, dtype: float64

/tmp/ipython-input-154143843.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0

```
sns.barplot(x=survival_rates.index, y=survival_rates.values, palette="pastel")
```



Mean survival rates by gender and embarkation port:

	Sex	Embarked	Survived
0	female	C	0.876712
1	female	Q	0.750000
2	female	S	0.689655
3	male	C	0.305263
4	male	Q	0.073171
5	male	S	0.174603

